

DBMS -- Study Notes Index

Personal interview-prep notes. Format: Rule + Example + 40s script + Follow-ups.
All topics are interview-answer ready. Update this file when new notes are added.

Files in This Folder

File
Topics Covered
Status

01-KEYS-AND-CONSTRAINTS.md

Super, Candidate, Primary, Alternate, FK, UK, Composite + 6 Constraints
[OK] Done

02-NORMALIZATION.md

1NF, 2NF, 3NF, BCNF -- with examples and 40s scripts
[OK] Done

03-JOINS-AND-ER.md

All JOIN types, ON vs WHERE, ER model basics
[OK] Done

04-SQL-TRAPS.md

8 must-know SQL traps with Wrong/Correct/Follow-up
[OK] Done

05-DBMS-TYPES.md

Hierarchical, Network, Relational, Object-Oriented + ER Model clarification
[OK] Done

06-3-TIER-ARCHITECTURE.md

3-tier architecture, layers, schema, 2-tier vs 3-tier
[OK] Done

07-SQL-COMMANDS.md

DDL/DML/DCL/TCL -- all commands, DROP vs TRUNCATE vs DELETE
[OK] Done

08-ACID-AND-RELATIONAL-MODEL.md

ACID properties (with bank examples), Isolation levels + 3 anomalies, Relational model terms, Degree vs Cardinality
[OK] Done

09-ER-MODEL.md

ER Model components, 4 attribute types, 4 cardinalities, Integrity Constraints (Domain/Entity/Referential/Key)
[OK] Done

10-RELATIONAL-ALGEBRA.md

6 RA operations (σ , π), SQL structure, WHERE clause, SET ops (UNION/INTERSECT/MINUS)
[OK] Done

11-FUNCTIONAL-DEPENDENCY.md

FD definition, Trivial/Non-Trivial/Completely Non-Trivial, Armstrong's 3 Axioms
[OK] Done

12-AGGREGATE-FUNCTIONS-AND-TRIGGERS.md

COUNT/SUM/AVG/MIN/MAX with NULL traps, Triggers (BEFORE/AFTER), AVG trap
[OK] Done

Revision Order (Follow This Sequence)

This is the ebook-aligned order for revision. Go through files in this sequence -- each one builds on the previous.

- Step 01 05-DBMS-TYPES.md DBMS types (Hierarchical/Network/Relational/OO)
- Step 02 06-3-TIER-ARCHITECTURE.md 3-tier architecture + schema
- Step 03 07-SQL-COMMANDS.md DDL/DML/DCL/TCL + DROP vs TRUNCATE vs DELETE
- Step 04 08-ACID-AND-RELATIONAL-MODEL.md ACID + Isolation anomalies + Relational terms
- Step 05 09-ER-MODEL.md ER Model + Integrity constraints
- Step 06 01-KEYS-AND-CONSTRAINTS.md All key types + SQL examples
- Step 07 10-RELATIONAL-ALGEBRA.md 6 RA ops + SQL structure + SET operations
- Step 08 03-JOINS-AND-ER.md JOIN types + ON vs WHERE + worked examples
- Step 09 04-SQL-TRAPS.md 8 SQL traps (do these AFTER JOINS)
- Step 10 11-FUNCTIONAL-DEPENDENCY.md FD types + Armstrong's axioms
- Step 11 02-NORMALIZATION.md Anomalies + Decomposition + 1NF 3NF + BCNF
- Step 12 12-AGGREGATE-FUNCTIONS-AND-TRIGGERS.md Aggregates + Triggers

Quick revision (day before interview): Steps 03, 04, 08, 09, 11 only.

Full revision (2-3 days out): All 12 steps in order.

Quick Cheat -- One Line Per Topic

- Keys Super Candidate {Primary, Alternate}; FK links tables; UK allows one NULL
- Constraints NOT NULL, UNIQUE, PK, FK, CHECK, DEFAULT
- 1NF atomic values, no repeating groups
- 2NF no partial dependency (non-key depends on FULL composite PK)
- 3NF no transitive dependency (non-key depends on non-key)
- JOINS INNER=match, LEFT=all left, RIGHT=all right, FULL=all both, CROSS=cartesian
- ER Model Entity(rect), Attribute(oval), Relationship(diamond); Cardinality 1:1/1:N/M:N
- SQL Trap 1 COUNT(*) counts NULLs; COUNT(col) skips NULLs
- SQL Trap 2 WHERE on right table kills LEFT JOIN move to ON clause
- SQL Trap 3 NOT IN + NULL subquery = empty result; use NOT EXISTS
- SQL Trap 4 WHERE before GROUP BY; HAVING after aggregation
- SQL Trap 5 NULL = NULL is UNKNOWN; use IS NULL
- SQL Trap 6 1-to-many JOIN inflates SUM/COUNT; pre-aggregate first
- SQL Trap 7 UNION deduplicates (slow); UNION ALL keeps dupes (fast)
- SQL Trap 8 AND > OR precedence; always parenthesize mixed conditions
- DBMS Types Hierarchical(tree), Network(graph), Relational(tables), OO(objects)
- 3-Tier Client(UI) App Server(logic) DB; no direct client-DB communication
- DDL CREATE, ALTER, DROP, TRUNCATE changes structure auto-committed
- DML SELECT, INSERT, UPDATE, DELETE changes data can rollback
- DCL GRANT, REVOKE controls access
- TCL COMMIT, ROLLBACK, SAVEPOINT controls transactions
- DROP removes table+structure (DDL, no rollback)
- TRUNCATE removes all rows, keeps structure (DDL, no rollback, fast)
- DELETE removes specific rows (DML, can rollback, fires triggers)
- ACID Atomicity(all-or-nothing), Consistency(valid valid), Isolation(no interference), Durability(survives crash)
- Isolation 3 anomalies: Dirty Read, Non-Repeatable Read, Phantom Read
4 levels: Read Uncommitted < Read Committed < Repeatable Read < Serializable(strictest)
- Degree number of columns in a table
- Cardinality number of rows in a table (different from ER cardinality!)
- ER Model Entity(rect), Attribute(oval), Relationship(diamond) -- design blueprint, NOT a DBMS type
- Attributes Simple, Key(underlined), Composite(sub-ovals), Multivalued(double oval), Derived(dashed oval)
- Cardinality 1:1 (Passport), 1:M (Dept Emp), M:1 (reverse), M:M (Emp Project via junction table)
- Integrity Domain(valid values), Entity(PK not null), Referential(FK matches PK), Key(PK unique)
- Surrogate artificial auto-generated key (auto-increment id); stable, not from business data
- Compound same as Composite key (2+ columns together form the key)
- RA ops sigma(WHERE/rows), pi(SELECT/cols), (UNION), (EXCEPT), (CROSS JOIN), (AS alias)
- SQL struct FROM(cartesian), WHERE(select predicate), SELECT(project)
- SET ops UNION(dedup), UNION ALL(keep dupes), INTERSECT(common), MINUS/EXCEPT(A not B)
- FD X Y means X determines Y; Trivial(Y X), Non-Trivial(Y X), Completely NT(X Y=)
- Armstrong Reflexivity(Y X X Y), Augmentation(X Y XZ YZ), Transitivity(X Y, Y Z X Z)
- Aggregates COUNT(*all), COUNT(col/no null), SUM, AVG(divides by non-NULLs!), MIN, MAX
- Trigger auto-fires on INSERT/UPDATE/DELETE; BEFORE(validate) or AFTER(audit/log)
- Anomalies Update(inconsistent copies), Delete(accidental loss), Insert(forced dummy data)
- Decompose Lossless(rejoin = original) + Dependency Preserving(FDs checkable in sub-tables)

Topics To Add Next

- [x] ER Model -- entities, attributes, relationships, weak entity, cardinality, integrity constraints ([OK] done in 09)
- [x] Relational Algebra -- all 6 operations with examples ([OK] done in 10)
- [x] Functional Dependency -- trivial vs non-trivial, Armstrong's Axioms ([OK] done in 11)
- [x] Aggregate Functions + Triggers ([OK] done in 12)

- [] Indexing -- B-tree vs Hash, clustered vs non-clustered
- [] Concurrency control -- locks, deadlock